

KAREN SIENA VILLANCIO-WOLTER

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EDUCATION

UNIVERSITY OF WASHINGTON

Master of Science in Electrical and Computer Engineering

NSF GRFP Fellow, College of Engineering Dean's Fellow

PhD Program in Electrical and Computer Engineering

Advanced to PhD Candidacy

Seattle, WA

Sep 2024-Jun 2026

UNIVERSITY OF FLORIDA

Bachelor of Science in Biomedical Engineering, summa cum laude

National Merit Scholar, Benacquisto Scholar, University Research Scholar

Gainesville, FL

Jul 2020-May 2024

RESEARCH EXPERIENCE

Ingraham Lab, Dr. Kim Ingraham

Graduate Researcher

Seattle, WA

Sep 2024-Present

- Designed 21 statistical models to analyze the effects of exoskeleton control parameters on user outcomes (energy cost, ankle kinematics, stride time) to optimize walking assistance for multiple objective functions simultaneously – paper in progress
- Built and tested a robotic hip exoskeleton; 3D-printed hardware, constructed circuits, designed electrical housing, added soft interface for user comfort, implemented structural improvements that eliminated two failure points
- Implementing a novel myoelectric exoskeleton controller; integrating EMG sensors, writing control code, modifying hardware
- Running user focus groups with 6 individuals with spinal cord injury to identify preferred exoskeleton interfaces and features

Human Neuromechanics Lab, Dr. Daniel P. Ferris

Undergraduate Researcher

Gainesville, FL

Sep 2021-May 2024

- Designed and led a study evaluating inverse kinematics pipelines using IMU sensors and motion capture data, identified that IMU heel placement resulted in joint angle calculations more similar to motion capture (standard) than dorsal placement
- Conducted 180+ hours of EMG-controlled exoskeleton data collections and post-stroke rehabilitation neuromechanics data collections (motion capture, EMG, IMUs, loadsol, COSMED, EEG, bodyweight support system, Bertec treadmill)

WORK EXPERIENCE

Medtronic Inc - Neurovascular Division

Manufacturing Engineering Intern

Irvine, CA

Jun-Aug 2023

- Co-led two DMAIC A3 improvement projects that reduced Solitaire (Revascularization Stent) reject rate from 20.2% to 9.6%
- Piloted a Gauge R&R study to investigate root causes of a prevalent rework code; trained and worked with manufacturing operators, maintained technical documentation, and presented study results to leadership
- Decreased frequency of misalignment reject code from 25% to 13% after installing new equipment in manufacturing line
- Calculated weekly product yield and material usage variance (MUV) reports, analyzed trends against previous weeks, led weekly product line update meetings with manufacturing engineering team and leadership
- Designed and prototyped a 3D-printed clean room fixture to address stent entanglement

Hanger Clinic - Prosthetics And Orthotics

Clinical Intern

Gainesville, FL

May-Aug 2022

- Built and adjusted prosthetic limbs using plaster molding, plastic pulling, and carbon fiber lamination
- Shadowed prosthetists and occupational therapists during patient visits for prosthetic arms, legs, and custom braces

LEADERSHIP/INVOLVEMENT

UF Generational Relief in Prosthetics (GRiP)

Research Project Manager, Pawsthetics Design Team Captain

Gainesville, FL

Jan 2021-May 2024

- Led a team of 10-15 students designing and building 3D-printed prosthetic limbs for 3 dogs
- Secured \$2,000 in project funding from the Donovan J. Welch Fellowship
- Ran biweekly meetings with 7 design team captains to monitor project progress and provide feedback and resources

PUBLICATIONS

Hybart, R., Villancio-Wolter, K. S., & Ferris, D. P. (2023). Metabolic cost of walking with electromechanical ankle exoskeletons ... PeerJ.

SKILLS/CERTIFICATIONS

Software: Python, MATLAB, C++, Javascript, HTML, CAD, PyTorch, Motion Capture (OpenSim, Visual3D, Motive), KiCad, Microsoft Office

Sensors/Electronics: Circuit Building, Machine Shop Tools, IMUs, EMG, EEG, ECG, Force Plates/Loadsol/FSRs, COSMED

Lean Six Sigma: DMAIC, SIPOC, FTQ, Fishbone

SolidWorks Certification: Associate - Mechanical Design